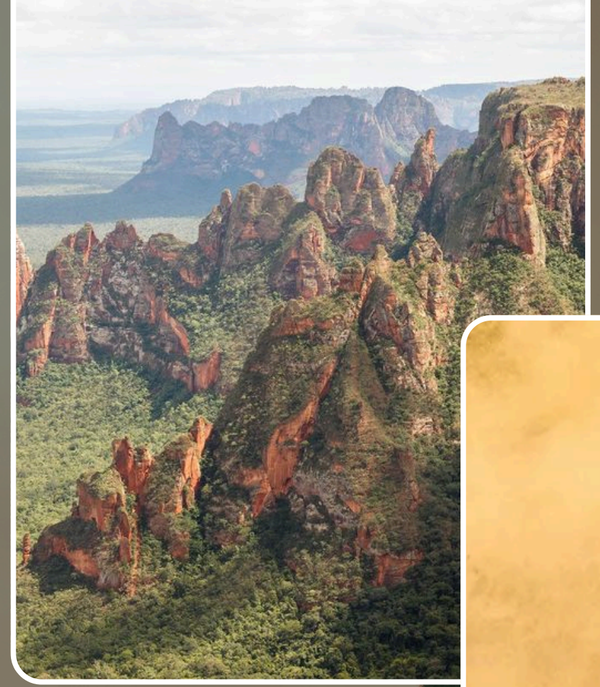
MANOUBA SCHOOL OF ENGINEERING

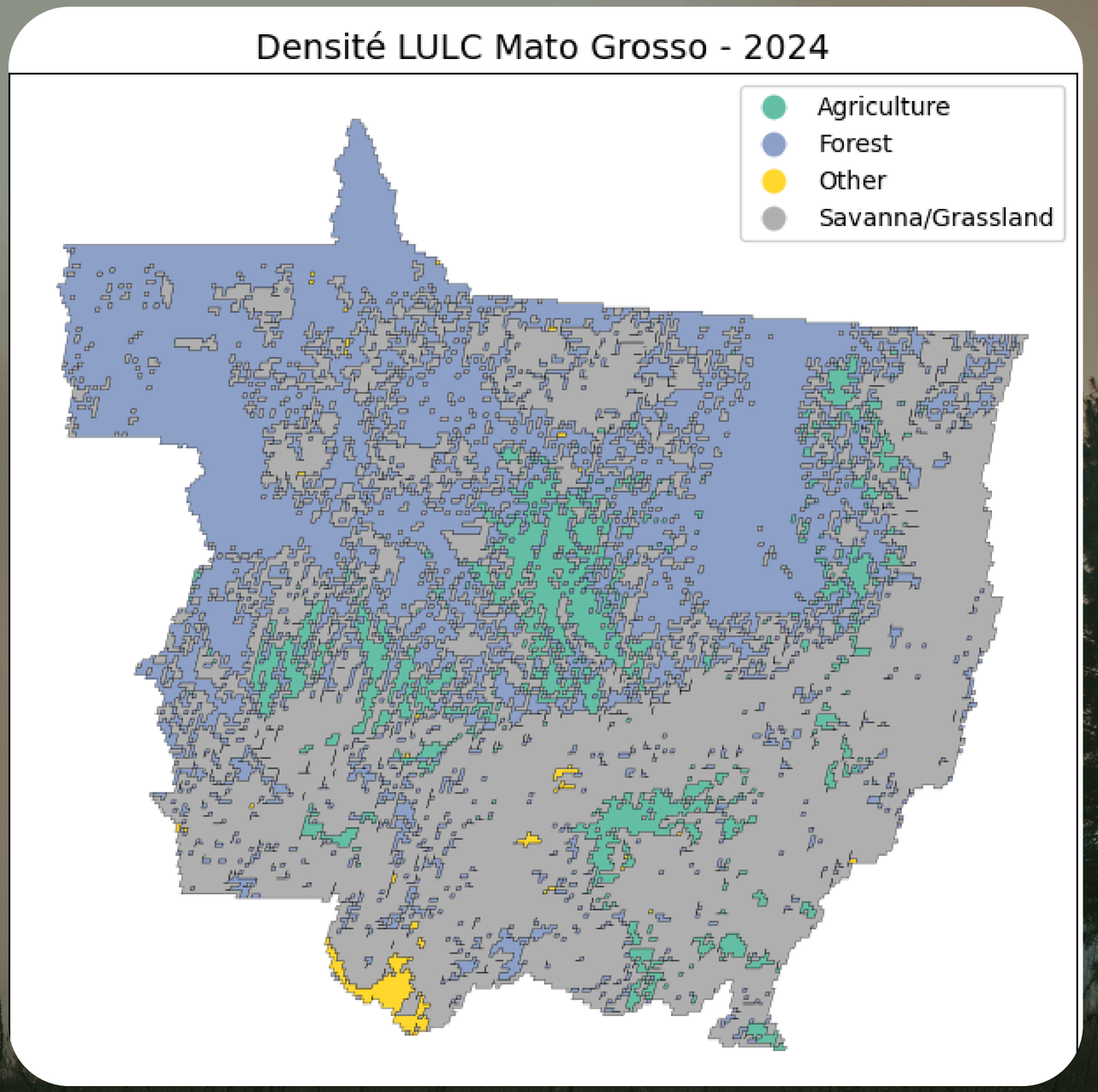
Exploring the

AMAZON FIRE CRISIS IN MATO GROSSO

REGION OF INTEREST



Mato Grosso, one of Brazil’s largest and most biodiverse states, is a critical hotspot where agricultural expansion, drought, and seasonal fires create an urgent need for advanced wildfire monitoring and prevention solutions.


Densité LULC Mato Grosso - 2024
Agriculture
Forest
Other
Savanna/Grassland

PROBLEM BEHIND OUR WORK

Rapid Fire Spread Across Forest and Agricultural Land

Environmental Damage and Biodiversity Loss

Risk to Communities, Health, and Economy

METHODOLOGY

step 1: study factors affecting fire

- collecting and processing Human activity data related to fire
- collecting and processing climate data related to fire(rainfall / wind/ temperature / drought...)
- collecting and processing Vegetation data related to fire

step 2: handling fire data

step 3: vizualization of data

step 4: Combining data collected to develop an algorithm for fire monitoring.

DATASET/POSDTGRES :COLLECTION /CLEANING /ORGANIZING

DATA HANDLING WORKFLOW

Data Ingestion: Multi-source integration 2019-2026

NASA FIRMS: Thermal hotspots (MODIS/VIIRS).
ERA5 (Copernicus): Climate variables (VPD, Temperature, Humidity).
Dynamic World: 10m resolution Land Use/Land Cover (LULC).
GEDI/NASA: Above Ground Biomass (AGB) density.

Spatial Normalization

Converting raw points and rasters into a unified PostgreSQL/PostGIS database.

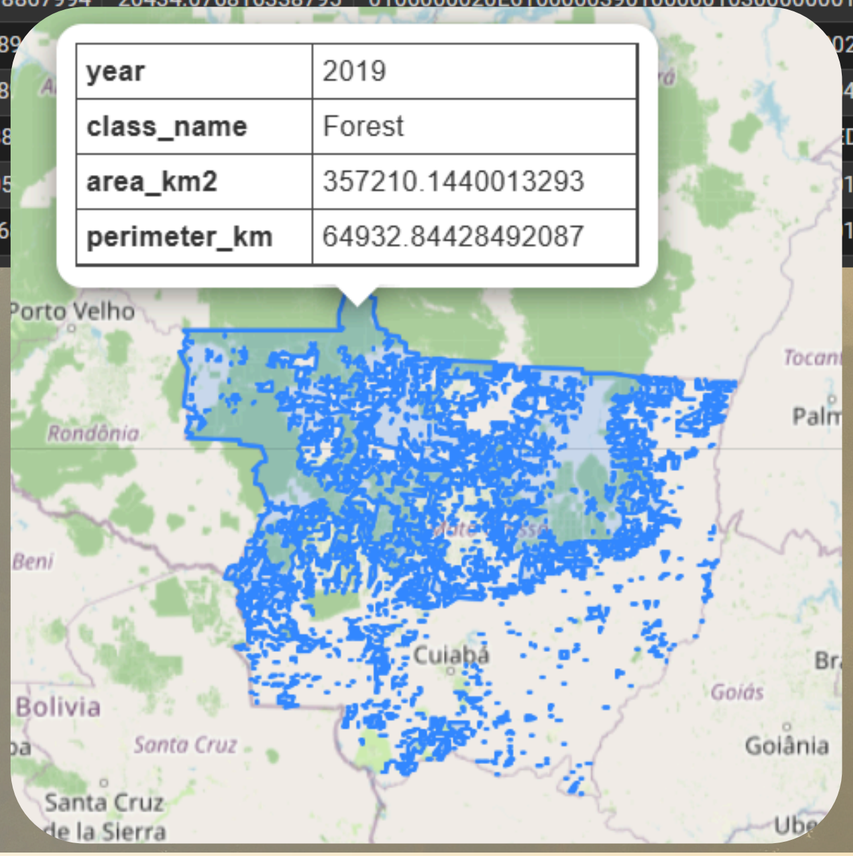
Index Calculation

On-the-fly calculation of NBR (Burn Ratio) and NDVI (Vegetation Health)

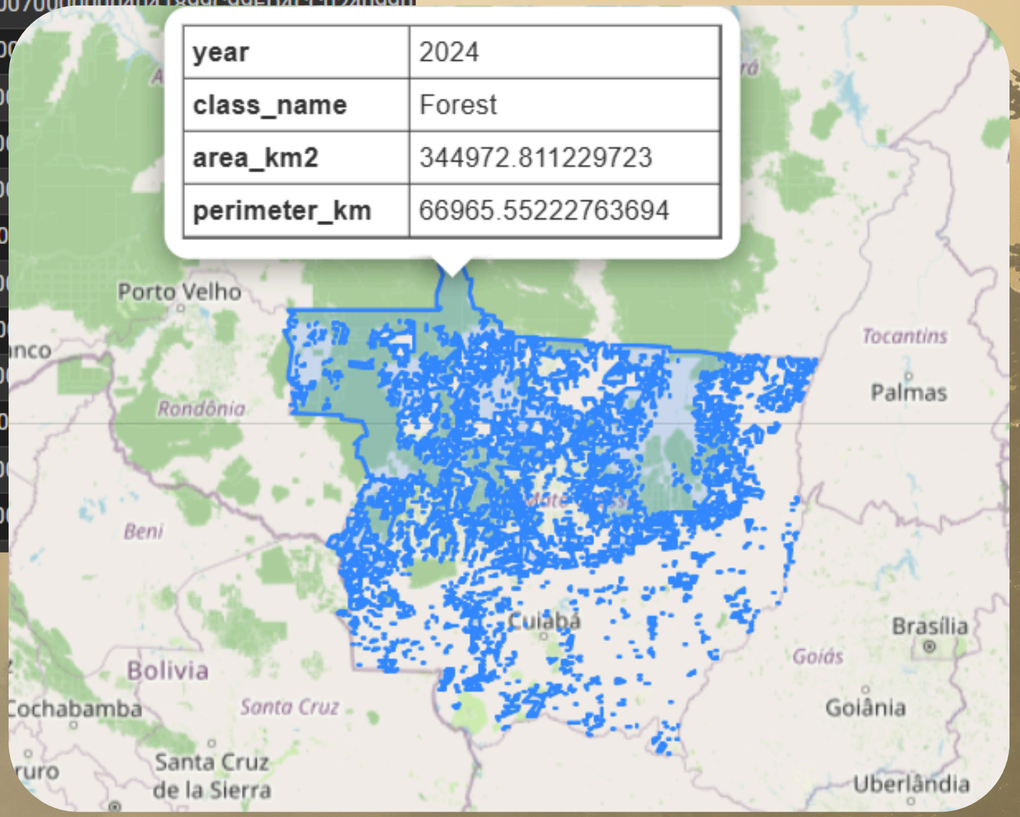
Showing rows: 1 to 24 | Page No: 1 of 1

	year bigint	class_name text	area_km2 double precision	perimeter_km double precision	geometry geometry
1	2019	Agriculture	61077.46969753364	19567.13224734067	0106000020E61000003A0100000103000000010000000500000009B21CA03D2154DC041CF63
2	2019	Forest	357210.1440013293	64932.84428492087	0106000020E6100000820300000103000000010000000500000001387A4AB98084EC0027DD1
3	2019	Other	8517.056302155906	2216.928435736873	0106000020E61000002D0000000103000000040000007000000040418899C99ED4CC0240990
4	2019	Savanna/Grassla...	478705.1399205863	81814.46211586607	0106000020E6100000070400000103000000C5010000
5	2020	Agriculture	62351.99623978905	20110.606059345984	0106000020E6100000420100000103000000010000000
6	2020	Forest	360033.6889331747	65713.14126262542	0106000020E6100000970300000103000000010000000
7	2020	Other	7471.336904632394	2013.0531217011837	0106000020E6100000250000000103000000030000000
8	2020	Savanna/Grassla...	475654.1197911905	82068.78304811184	0106000020E61000001E0400000103000000DB01000
9	2021	Agriculture	64076.601018867994	20434.076816338795	0106000020E6100000390100000103000000010000000
10	2021	Forest	355591.87789		0200000
11	2021	Other	6798.854678		4000000
12	2021	Savanna/Grassla...	479041.45488		ED01000
13	2022	Agriculture	62731.776105		1000000
14	2022	Forest	346118.0466		1000000

year	2019
class_name	Forest
area_km2	357210.1440013293
perimeter_km	64932.84428492087



year	2024
class_name	Forest
area_km2	344972.811229723
perimeter_km	66965.55222763694



Data Output | Messages | Geometry Viewer | Graph Visualiser | Notifications

Showing rows: 1 to 48 | Page No: 1 of 1

	year bigint	class_name text	area_km2 double precision	perimeter_km double precision	geometry geometry
1	2019	Bare Ground	6075.000000524125	4530.000000200506	0106000020E6100000D400000001030000000100000005000000 8074C80CA7554CC06F0F79
2	2019	Built-Up Area	1125.0000000451296	710.0000000138708	0106000020E61000001A000000010300000001000000050000001 57534405CD94CC0CA56A0
3	2019	Crops	84150.00000442527	39820.00000132554	0106000020E61000002D040000010300000001000000050000009 C2CDB046E1B4DC06732FE
4	2019	Flooded Vegetati...	1724.9999999628642	949.9999999825817	0106000020E6100000200000000103000000010000000500000 AEB3AF780DD44CC037E7C1
5	2019	Grass	128950.00001362716	70580.00000393047	0106000020E6100000CA080000010300000001000000050000001 1D01BD9F1C24DC086699F
6	2019	Shrub & Scrub	101900.00002400769	5	
7	2019	Trees	575825.0000514637	1	
8	2019	Water	4925.000000480601	3	
9	2020	Bare Ground	6446.753222328034		
10	2020	Built-Up Area	1069.3594117485022		
11	2020	Crops	95060.96919547525		
12	2020	Flooded Vegetati...	568.462216840426		
13	2020	Grass	116931.91871957824		
14	2020	Shrub & Scrub	113658.99965081978		
15	2020	Trees	566634.9978865557		

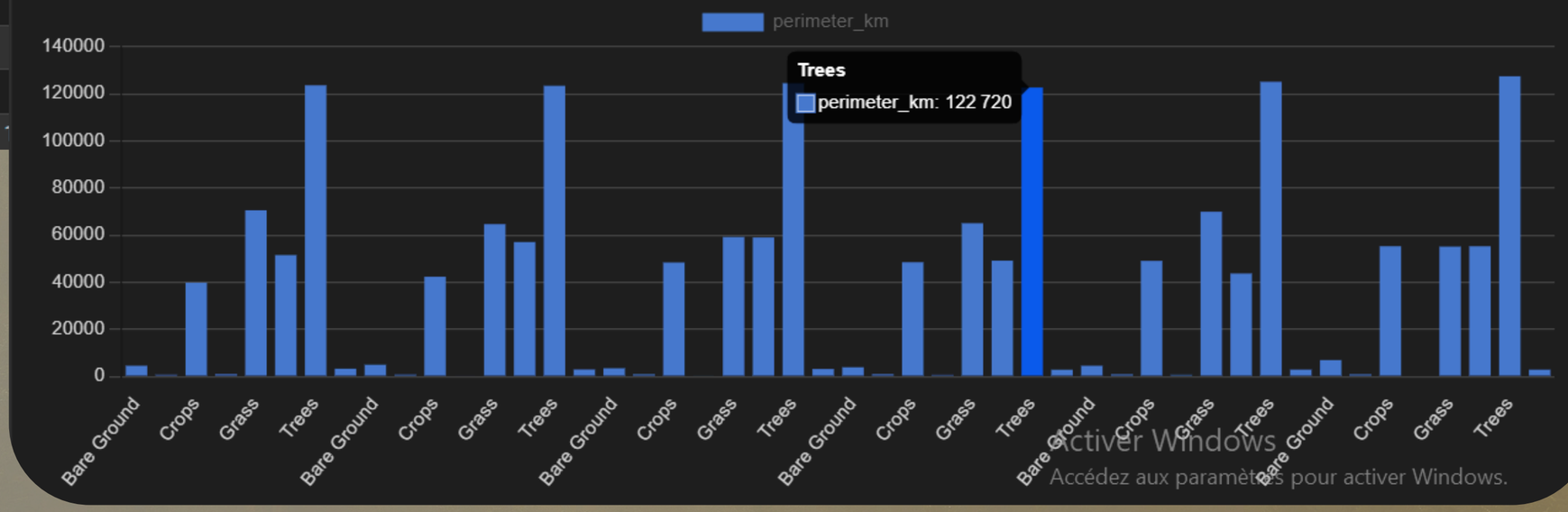
Data Output | Messages | Geometry Viewer | Graph Visualiser | Notifications

Graph Type: Bar Chart

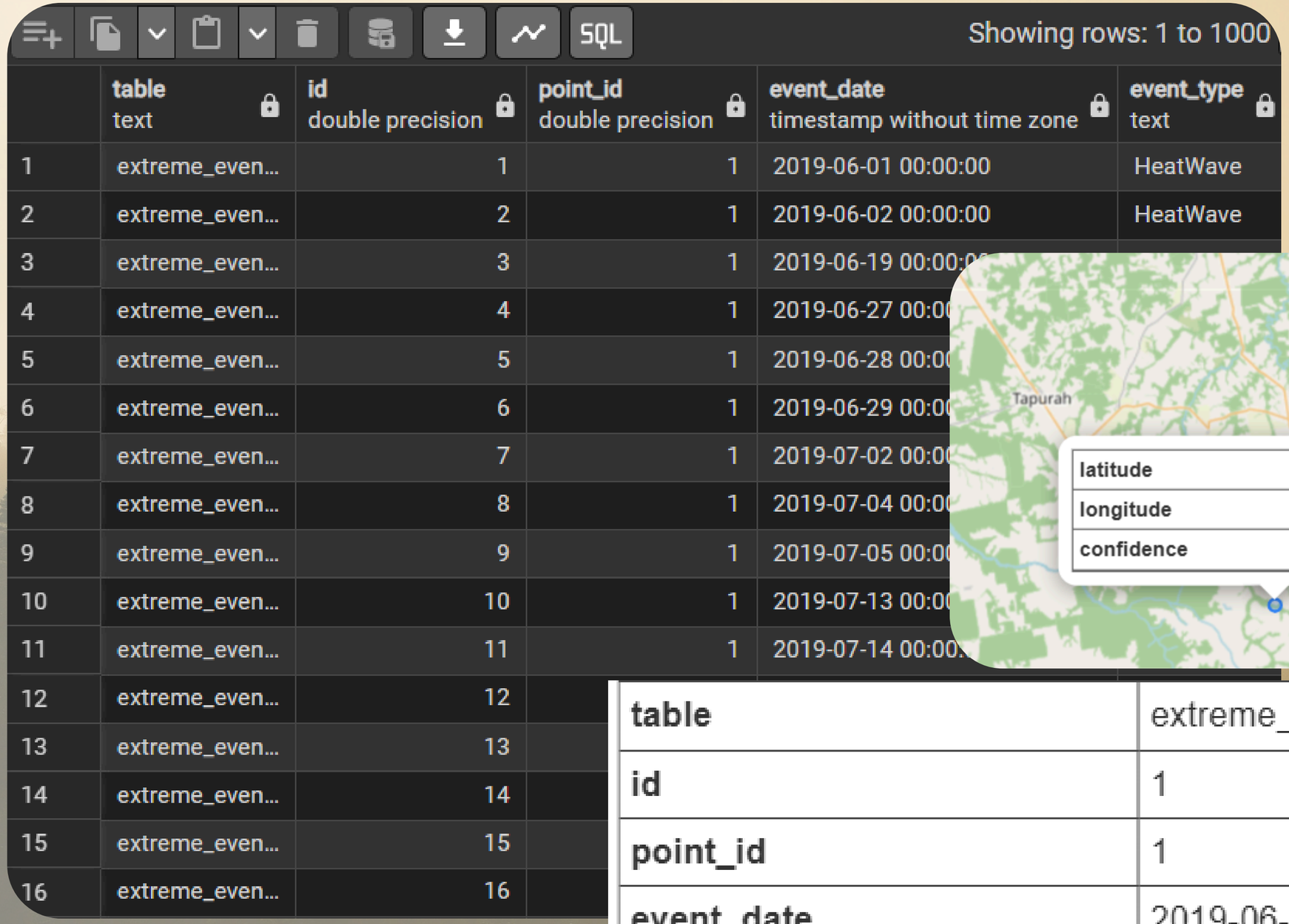
X Axis: class_name

Y Axis: perimeter_km

Generate



Activer Windows
Accédez aux paramètres pour activer Windows.



Showing rows: 1 to 1000

	table text	id double precision	point_id double precision	event_date timestamp without time zone	event_type text
1	extreme_even...	1	1	2019-06-01 00:00:00	HeatWave
2	extreme_even...	2	1	2019-06-02 00:00:00	HeatWave
3	extreme_even...	3	1	2019-06-19 00:00:0	
4	extreme_even...	4	1	2019-06-27 00:0	
5	extreme_even...	5	1	2019-06-28 00:0	
6	extreme_even...	6	1	2019-06-29 00:0	
7	extreme_even...	7	1	2019-07-02 00:0	
8	extreme_even...	8	1	2019-07-04 00:0	
9	extreme_even...	9	1	2019-07-05 00:0	
10	extreme_even...	10	1	2019-07-13 00:0	
11	extreme_even...	11	1	2019-07-14 00:0	
12	extreme_even...	12			
13	extreme_even...	13			
14	extreme_even...	14			
15	extreme_even...	15			
16	extreme_even...	16			

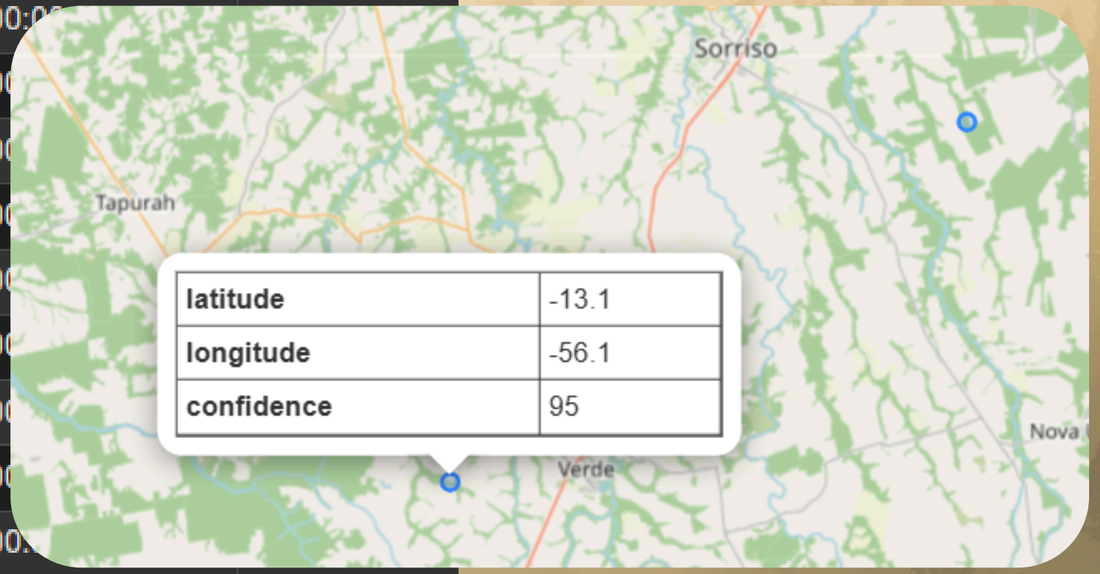


table	extreme_even
id	1
point_id	1
event_date	2019-06-01 0(
event_type	HeatWave

created_at timestamp without time zone	municipality text	state text	region text	geometry geometry
2026-04-15 21:41:15.23	Sinop	Mato Gros…	Amazon	0101000020E6100000000000000804BC00000000000002…
2026-04-15 21:41:15.23	Sinop	Mato Gros…	Amazon	0101000020E6100000000000000804BC00000000000002…
2026-04-15 21:41:15.23	Sinop	Mato Gros…	Amazon	0101000020E6100000000000000804BC00000000000002…
2026-04-15 21:41:15.23	Sinop	Mato Gros…	Amazon	0101000020E6100000000000000804BC00000000000002…
2026-04-15 21:41:15.23	Sinop	Mato Gros…	Amazon	0101000020E6100000000000000804BC00000000000002…
2026-04-15 21:41:15.23	Sinop	Mato Gros…	Amazon	0101000020E6100000000000000804BC00000000000002…
2026-04-15 21:41:15.23	Sinop	Mato Gros…	Amazon	0101000020E6100000000000000804BC00000000000002…
2026-04-15 21:41:15.23	Sinop	Mato Gros…	Amazon	0101000020E6100000000000000804BC00000000000002…
2026-04-15 21:41:15.23	Sinop	Mato Gros…	Amazon	0101000020E6100000000000000804BC00000000000002…
2026-04-15 21:41:15.23	Sinop	Mato Gros…	Amazon	0101000020E6100000000000000804BC00000000000002…
2026-04-15 21:41:15.23	Sinop	Mato Gros…	Amazon	0101000020E6100000000000000804BC00000000000002…
2026-04-15 21:41:15.23	Sinop	Mato Gros…	Amazon	0101000020E6100000000000000804BC00000000000002…
2026-04-15 21:41:15.23	Sinop	Mato Gros…		
2026-04-15 21:41:15.23	Sinop	Mato Gros…		
2026-04-15 21:41:15.23	Sinop	Mato Gros…		
2026-04-15 21:41:15.23	Sinop	Mato Gros…		

created_at	2026-04-15 21:41:15.23
municipality	Sinop
state	Mato Grosso
region	Amazon

NDVI double precision	agb double precision	dynamic_world_class double precision	total_precipitation_sum double precision	u_component_of_wind_10m double precision	v_component_of_wind_10m double precision
8439.739130434782	115.00910186767578	1	0.11090800745165492	-0.456021932047226	0.1033242020522659
8439.739130434782	115.00910186767578	1	0.11090800745165492	-0.456021932047226	0.1033242020522659
8439.739130434782	115.00910186767578	1	0.11090800745165492	-0.456021932047226	0.1033242020522659
8439.739130434782	115.00910186767578	1	0.11090800745165492	-0.456021932047226	0.1033242020522659
8439.739130434782	115.00910186767578	1	0.11090800745165492	-0.456021932047226	0.1033242020522659
8439.739130434782	115.00910186767578	1	0.11090800745165492	-0.456021932047226	0.1033242020522659
8439.739130434782	115.00910186767578	1	0.11090800745165492	-0.456021932047226	0.1033242020522659
8439.739130434782	115.00910186767578	1			
8439.739130434782	115.00910186767578	1			
8439.739130434782	115.00910186767578	1			
8439.739130434782	115.00910186767578	1			
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8439.739130434782	115.00910186767578	1			
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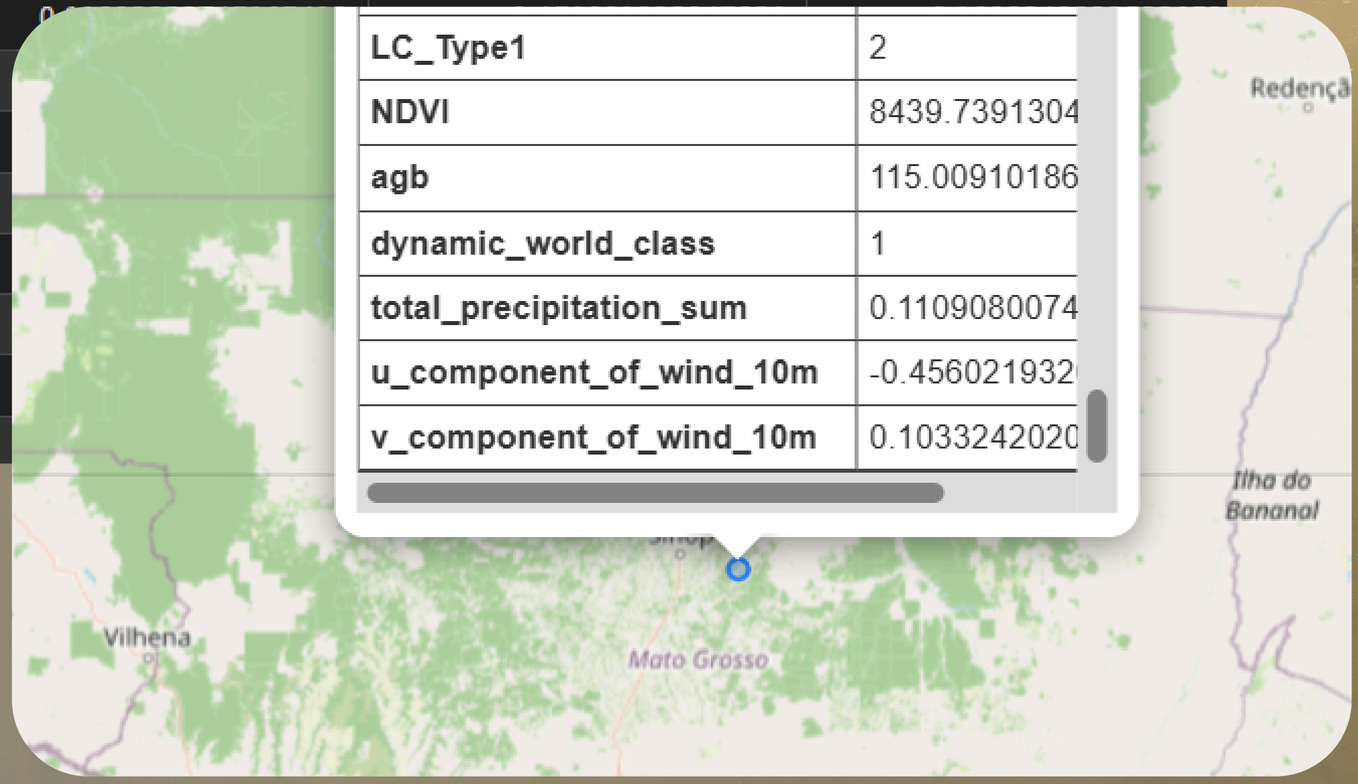

985
986
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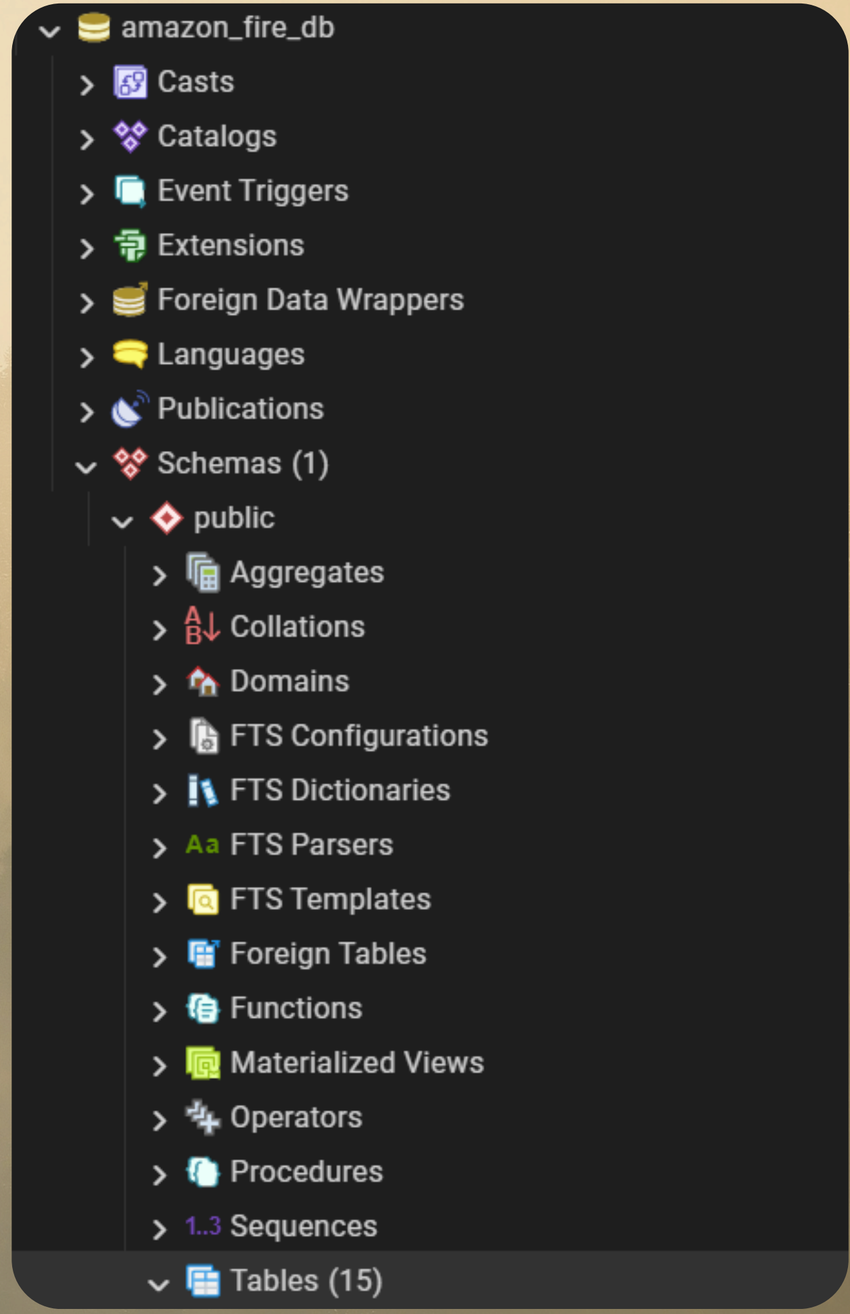
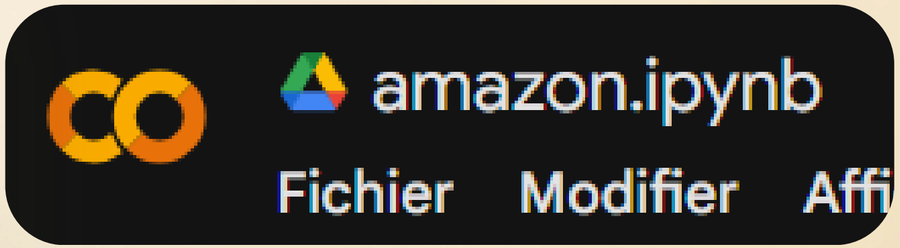
999
1000

```
Processing year 2019...
Sampling 3363 points from GEE...
Enriched 3363 points for 2019.
Processing year 2020...
Sampling 3419 points from GEE...
Enriched 3419 points for 2020.
Processing year 2021...
Sampling 2773 points from GEE...
Enriched 2773 points for 2021.
Processing year 2022...
Sampling 2564 points from GEE...
Enriched 2564 points for 2022.
Processing year 2023...
Sampling 5184 points from GEE...
Enriched 5184 points for 2023.
Processing year 2024...
Sampling 5145 points from GEE...
Enriched 5145 points for 2024.
Processing year 2025...
Sampling 4647 points from GEE...
```



DATA VISUALIZATION / MONITORING




```python
import numpy as np
import pandas as pd
from datetime import datetime, timedelta
```

Data manipulation

```python
import geopandas as gpd
from shapely.geometry import Point, Polygon
from shapely.ops import unary_union
import rasterio
from rasterio.mask import mask
import laspy
```

Spatial data

Remote sensing

```
import ee
import geemap
import rioxarray
import xarray as xr
```

Visualization

```
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import seaborn as sns
import folium
from folium import plugins
import plotly.express as px
import plotly.graph_objects as go
from plotly.subplots import make_subplots
```

Statistics

```
from scipy import stats
from scipy.spatial import distance_matrix
```

Time series

```
from statsmodels.tsa.seasonal import seasonal_decompose
from statsmodels.tsa.arima.model import ARIMA
from prophet import Prophet
```

notebook results style then integrated in the website

Analysis Notebook

- 01 Climate Analysis
- 02 Human Impact
- 03 Vegetation & Land Cover
- 04 Land Use Change
- 05 Loss Distribution
- 06 Risk Assessment
- 07 Final Synthesis

v2.4.1-stable
Mato Grosso Region

01 Climate Analysis

> Analysis of multi-year meteorological datasets reveals a significant thermal anomaly during the June-August window. Mean temperature trends show a distinct upward trajectory, inversely correlated with precipitation cycles. The drought index monitoring indicates a prolonged period of moisture deficit, increasing the susceptibility of the region to biomass ignition.

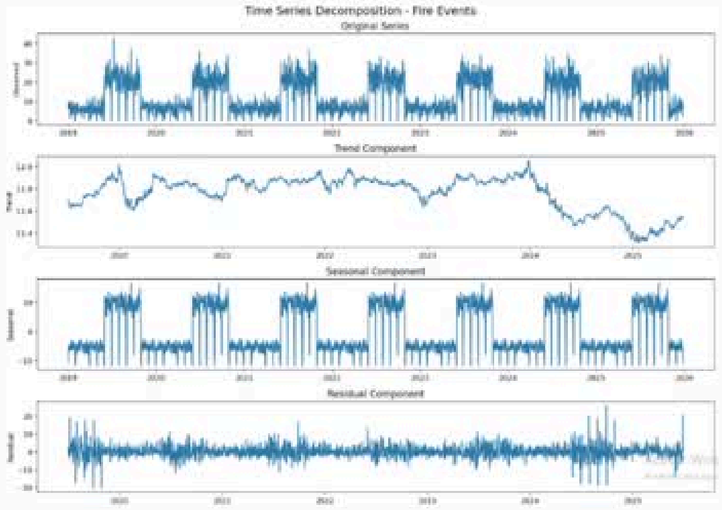


Figure 1.1: Multi-component time series decomposition (2019-2025) showing

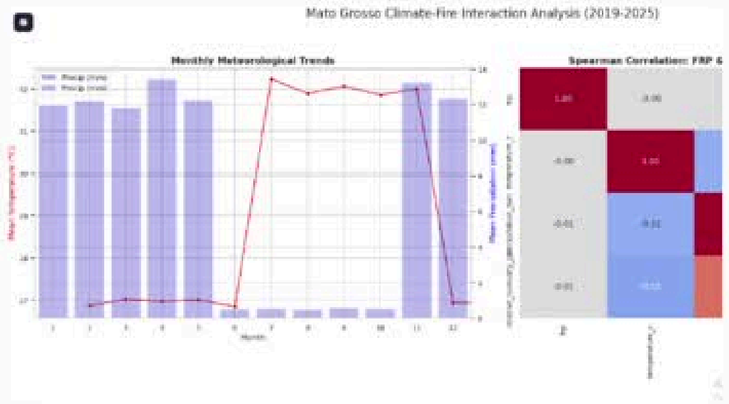
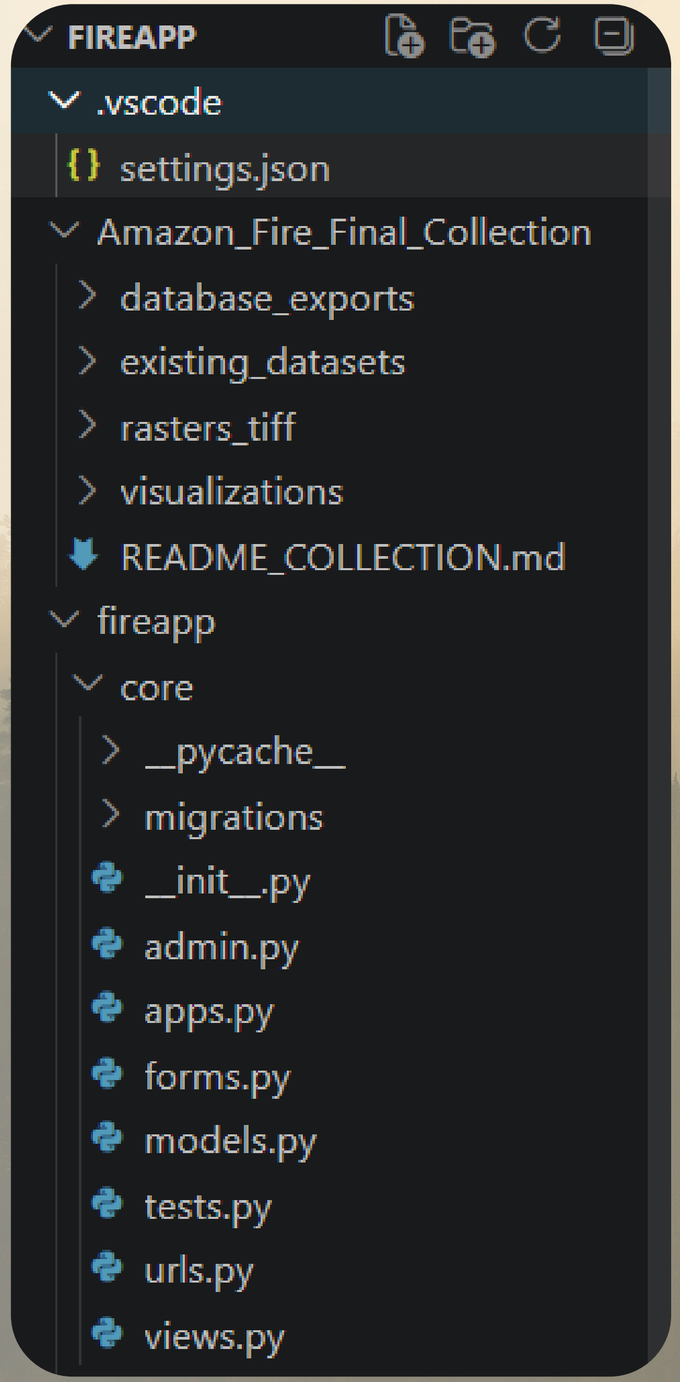
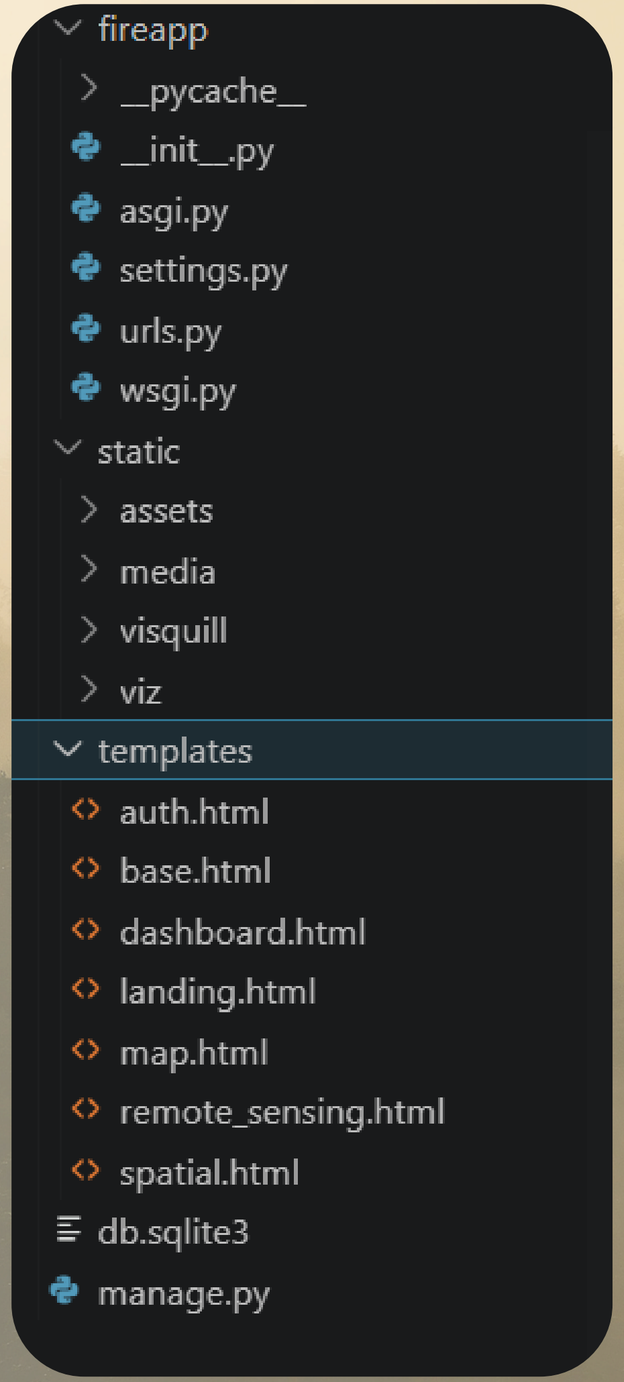


Figure 1.2: Monthly meteorological trends highlighting temperature and precipitation interaction

Activer Windows
Accédez aux paramètres pour activer Windows.

CODE/BACKEND





Project Architecture in vs code

The architecture follows the Django MVT (Model-View-Template) pattern, organized as follows:

Backend (The Logic):

Models: Handles data structures and the db.sqlite3 database.
Views: Processes spatial logic and connects the data to the interface.
Core: Contains the main application settings and routing (urls.py).

Frontend (The Interface):

Templates: Modular HTML files for the Dashboard, Map, and Remote Sensing views.

Static Assets: Stores CSS, JavaScript, and UI assets for the spatial dashboard.

Data Layer:
Rasters & Collections: Dedicated storage for GeoTIFFs, satellite imagery, and database exports used in analysis.

To run Django

1. Prerequisites

Ensure you have Python installed and the necessary
pip install django pandas

2. Navigate to the Project Folder

Open your terminal and navigate to the directory containing `manage.py`:
cd c:\Users\msi\Downloads\fireapp\fireapp

3. Initialize the Database

If this is the first time running the app, apply the migrations to set up the SQLite database:
python manage.py migrate

4. Start the Development Server

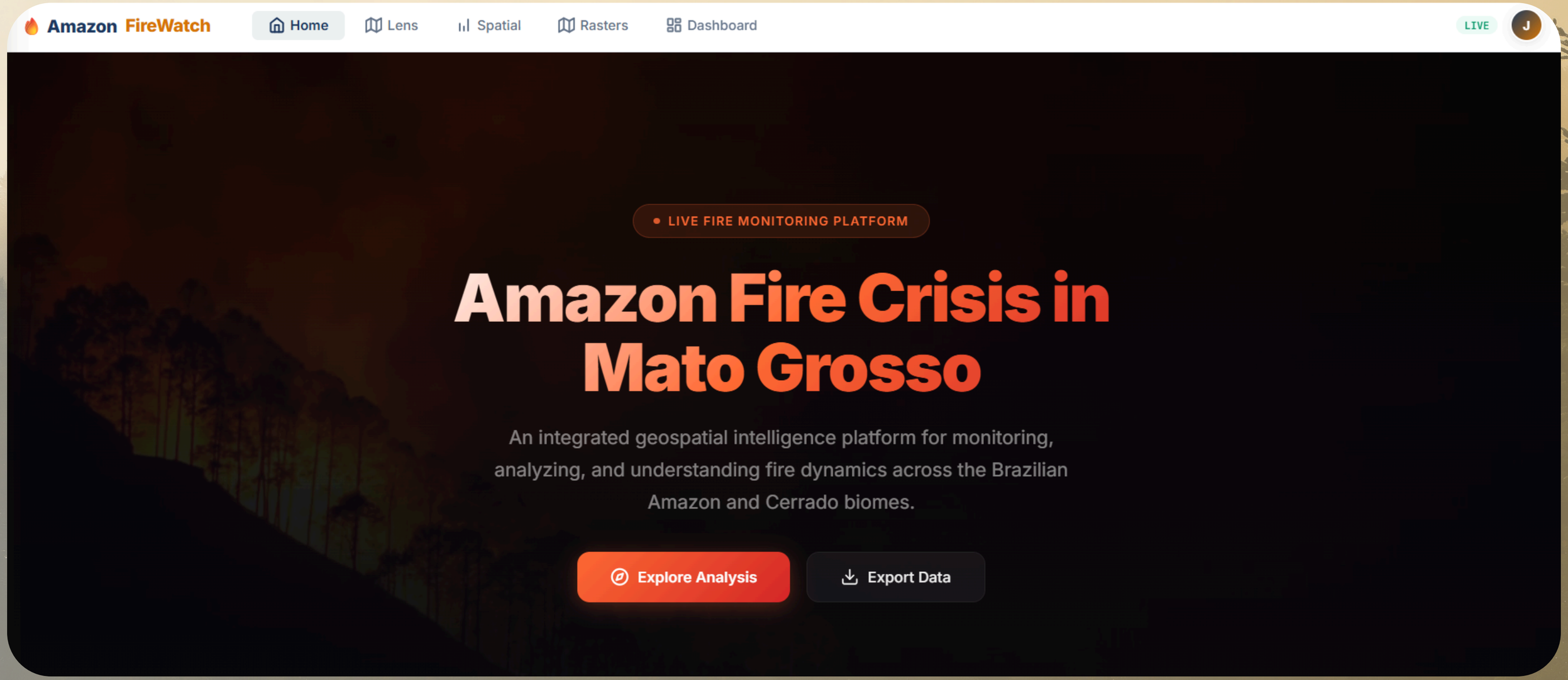
Run the following command to start the app:
python manage.py runserver

5. Access the App

Once the server is running, open your web browser and go to:
http://127.0.0.1:8000

SOME FEATURES TO UNDERSTAND THE FUNCTIONNALITY OF OUR WEBSITE

DATA HUB


Amazon FireWatch
Home
Lens
Spatial
Rasters
Dashboard
LIVE
J
LIVE FIRE MONITORING PLATFORM
Amazon Fire Crisis in Mato Grosso
An integrated geospatial intelligence platform for monitoring, analyzing, and understanding fire dynamics across the Brazilian Amazon and Cerrado biomes.
Explore Analysis
Export Data

Analysis Tools

Three powerful lenses into the fire crisis — from spatial mapping to temporal analysis

Interactive Fire Map

Explore fire hotspots across Mato Grosso with an advanced lens-based map viewer. Visualize fire density and event clusters.

LENS VIEWER

Spatia

Leaflet-powered a
cluster markers,
Explore real-tin

GIS A

Export Data

Download fire incident data for your own analysis

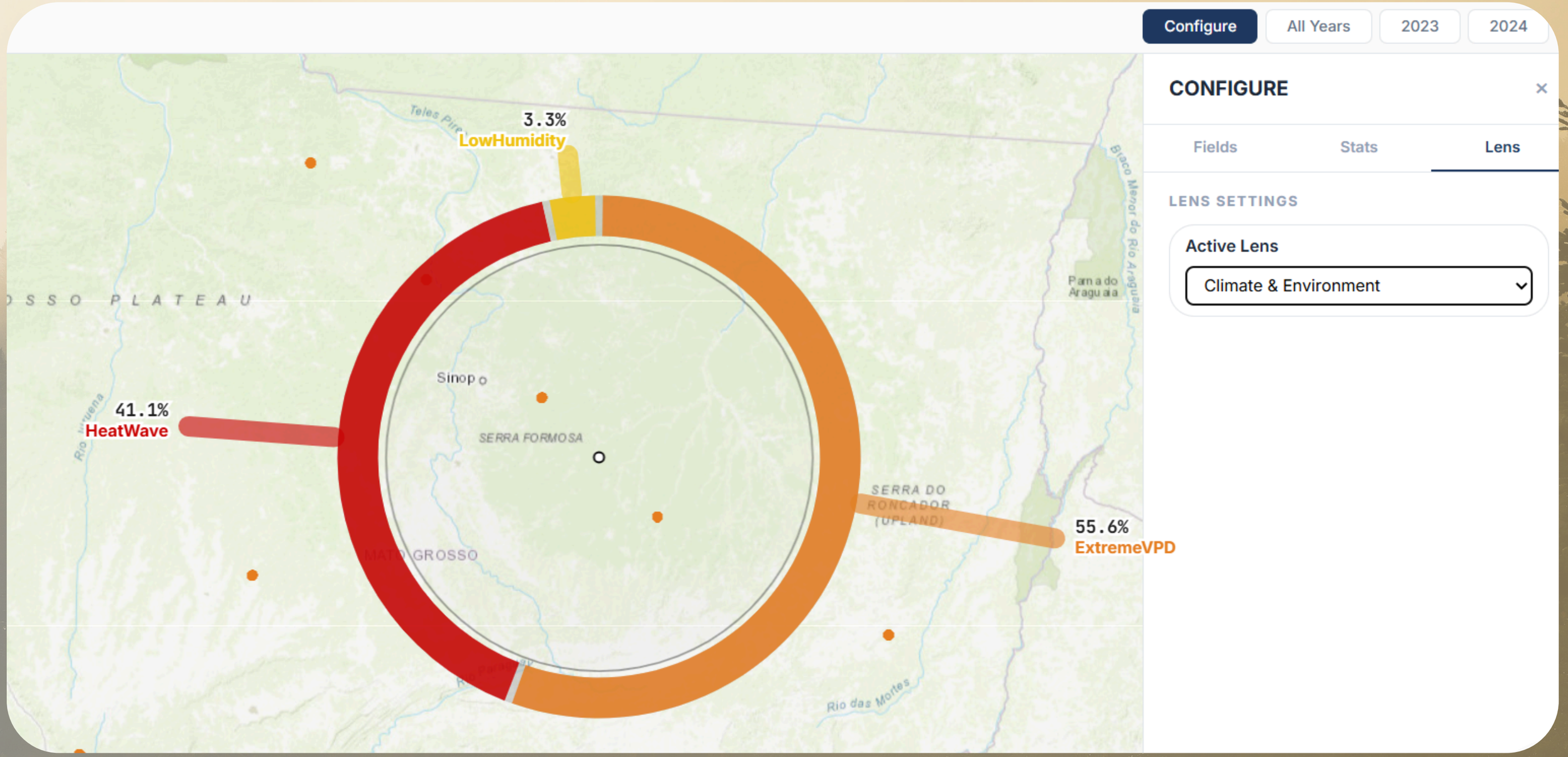
CSV Format

Tabular data for spreadsheets & statistics

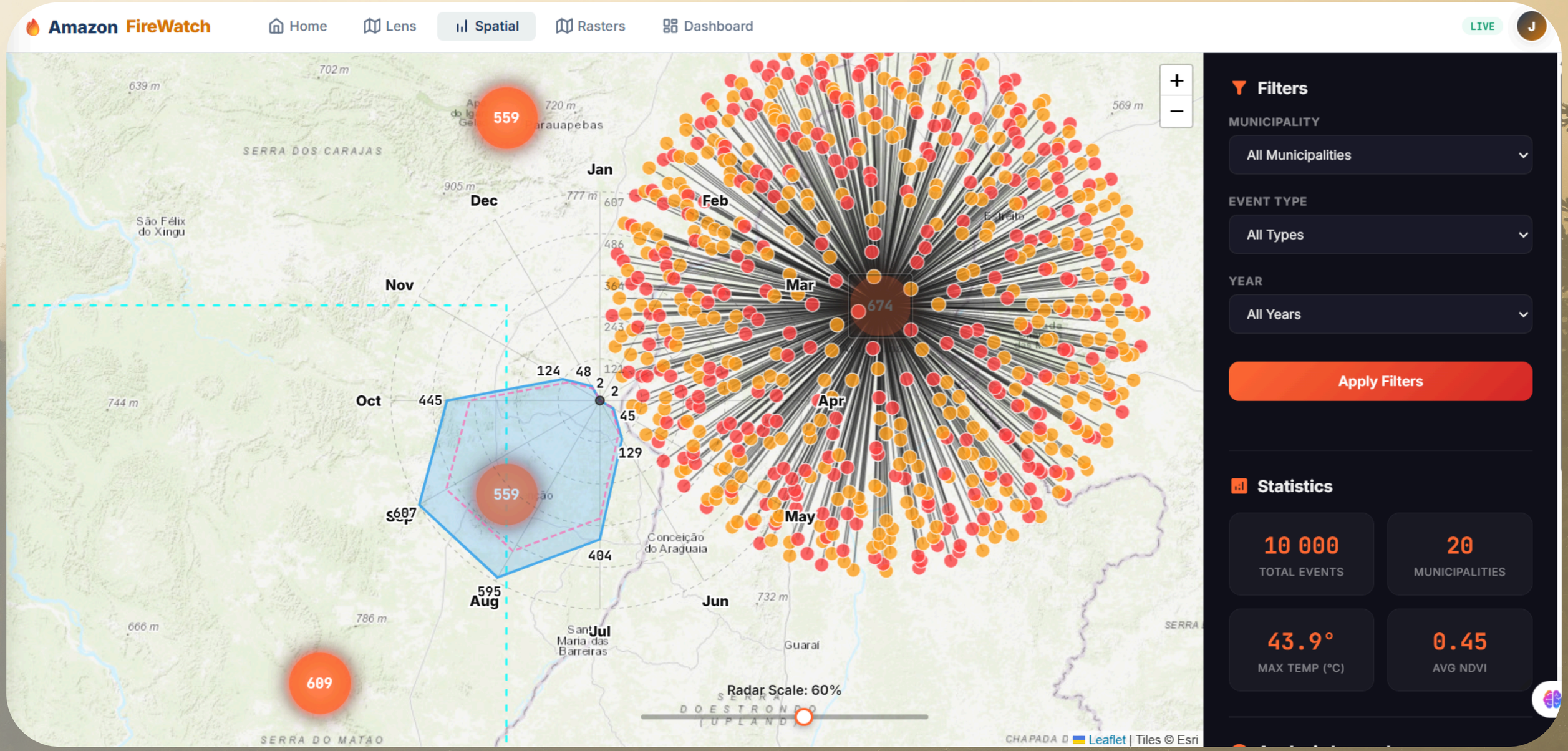
GeoJSON Format

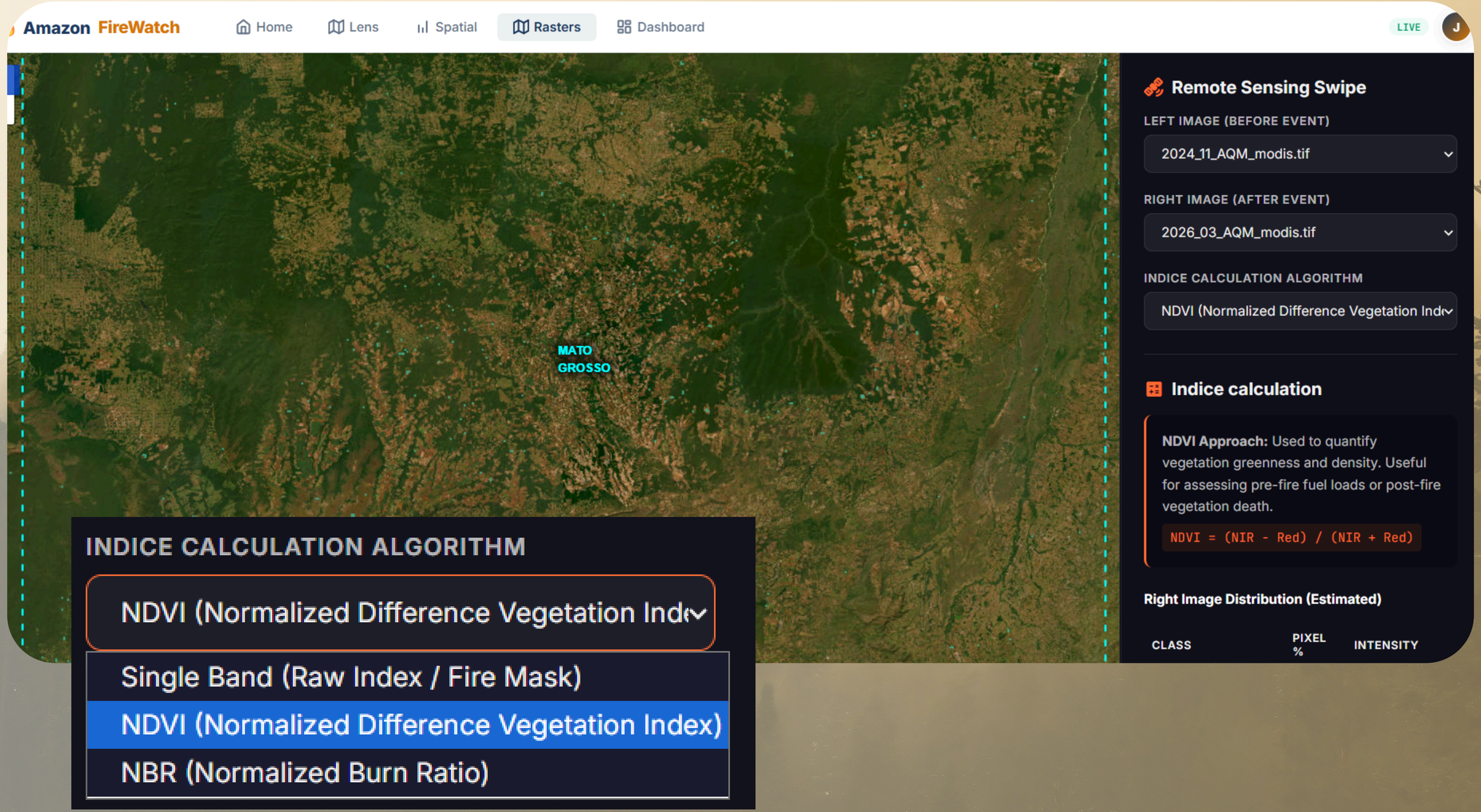
Geospatial data for GIS applications

original feature ↰



original feature 2




Amazon FireWatch
Home
Lens
Spatial
Rasters
Dashboard
LIVE
J
MATO
GROSSO
Remote Sensing Swipe
LEFT IMAGE (BEFORE EVENT)
2024_11_AQM_modis.tif
RIGHT IMAGE (AFTER EVENT)
2026_03_AQM_modis.tif
INDICE CALCULATION ALGORITHM
NDVI (Normalized Difference Vegetation Inde
Indice calculation
NDVI Approach: Used to quantify vegetation greenness and density. Useful for assessing pre-fire fuel loads or post-fire vegetation death.
NDVI = (NIR - Red) / (NIR + Red)
Right Image Distribution (Estimated)
CLASS
PIXEL %
INTENSITY
INDICE CALCULATION ALGORITHM
NDVI (Normalized Difference Vegetation Inde
Single Band (Raw Index / Fire Mask)
NDVI (Normalized Difference Vegetation Index)
NBR (Normalized Burn Ratio)

TECHNOLOGY STACK

Languages:

Python: Data engineering, statistics, and backend logic.

JavaScript (ES6+): Interactive mapping and client-side raster processing.

SQL (PostGIS): Spatial queries and relationship mapping.

Software & Frameworks:

Backend: Django (Python Web Framework).

Mapping: Leaflet.js (Core map engine) + GeoRaster (TIFF processing).

Visualization: Chart.js (Interactive dashboards).

Styling: Vanilla CSS (Glassmorphism design system).

Ingoude Company

Technical Constraints & Challenges

Browser-Side Processing

Processing large GeoTIFF files (Rasters) in the user's browser requires high memory optimization (using `georaster-layer`).

Data Heterogeneity

Harmonizing data from different sensors with varying spatial resolutions (10m for LULC vs 375m for VIIRS)

Performance

Maintaining 60fps animations on maps while handling 27,000+ fire incident markers simultaneously.

Future work
what's next

- Predictive AI: Integration of Machine Learning to forecast fire probability for the next 7 days based on current soil moisture and temperature.(action)

- Before and after swipe like gee

- Real-time API connection to NASA FIRMS for "Zero-Hour" monitoring.
- Fire spread simulation
- Financial loss estimation
- Enhance data treatment
- Add other factors like DEMs topography

LIVE DEMO

THANK YOU!